

Exception Handling Assignments

Exception Assignment 1: DuplicateCityException in Delivery Service

Concepts: HashSet, custom exception, validation

Level: Beginner

Problem Statement:

Extend the *Unique City Names for Delivery Service* application to handle duplicate entries using a custom exception.

- Define a custom exception class DuplicateCityException that is thrown when someone tries to add a city that already exists in the set.
- Modify the logic so that:
 - If a city already exists, the exception is thrown with a proper message (e.g., "City already in delivery list: Pune").
 - The exception is caught and a user-friendly message is shown.

Exception Assignment 2: EmployeeNotFoundException in Employee Directory

Concepts: HashMap, custom checked exception, key validation

Level: Intermediate

Problem Statement:

Enhance the *Employee Directory* assignment with error handling when searching for or deleting an employee.

- Create a custom checked exception called EmployeeNotFoundException.
- This exception should be thrown when:
 - A fetch or delete operation is attempted using an employee ID that doesn't exist in the directory.
- Catch the exception and print a meaningful message (e.g., "No employee found with ID: 1034").

Exception Assignment 3: GenreNotFoundException in Bookstore Catalog

Concepts: TreeMap, TreeSet, nested collections, custom exception, validation

Level: Intermediate

Problem Statement:

Update the *Bookstore Catalog* system to handle errors gracefully using a custom exception.

- Define a custom exception GenreNotFoundException which is thrown when:
 - A user tries to add/remove a book from a genre that doesn't exist
 - Or tries to list books from an invalid genre

- Catch the exception and show proper messages like “Genre 'Fantasy' does not exist in catalog.”